1. Work Breakdown Structure (WBS)

Phase 1: Data Acquisition & Staging (Capture)

- **1.1 DOT Airfare Ingestion:** Connect to the Data.gov Socrata API (data.transportation.gov) and pull quarterly domestic route records (city1, city2, nsmiles, passengers, fare, carrier_lg, large_ms, fare_lg).
- **1.2 EIA Jet Fuel Ingestion:** Query the U.S. EIA API for matching monthly/quarterly spot prices of jet fuel to control for external fuel cost spikes.
- **1.3 Raw Staging:** Save raw API output locally in structured JSON format (/data/raw/).

Phase 2: Transformation & Feature Engineering (Process & Store)

- **2.1 Data Cleaning & Filtering:** Strip currency symbols from fares (fare, fare_lg), convert large_ms to floating-point decimals, standardize city/state naming, and drop routes with < 10 passengers per day.
- **2.2 Enrichment Join:** Merge the EIA quarterly fuel price index with the DOT route records by matching year and quarter.
- **2.3 Feature Generation:**
    - Calculate dollar markup (fare_lg - fare) and percentage markup.
    - Categorize competition tiers: Competitive ($<40\%$), Concentrated ($40\%\text{–}65\%$), and Monopolized ($>65\%$).
    - Compute cost-per-mile (fare / nsmiles).
- **2.4 Partitioned Storage:** Write cleaned DataFrames into Apache Parquet partitioned by /data/processed/year=YYYY/quarter=Q/.

Phase 3: Modeling & Statistical Testing (Analyze)

- **3.1 Monopoly Surcharge Testing:** Run one-way ANOVA and independent two-sample t-tests across the three competition tiers to isolate the exact dollar markup while controlling for distance.
- **3.2 Distance Elasticity Analysis:** Measure correlation and perform scatter regression comparing flight distance to markup, assessing pricing power on short-haul flights (< 400 miles) versus transcontinental routes.
- **3.3 Fair-Fare Predictive Modeling:** Train a predictive model (Multiple Linear Regression / Random Forest) using distance, market share, passenger volume, quarter seasonality, and fuel price to establish an expected fair market fare.
- **3.4 Gouging Anomaly Detection:** Calculate residuals (actual_fare - predicted_fare) to identify and rank the top-10 most overpriced routes in the country.

Phase 4: Notebook Assembly & Code Quality (Deliverable 1)

- **4.1 Pipeline Integration:** Assemble Phase 1 through Phase 3 sequentially inside a single clean Jupyter Notebook (.ipynb).
- **4.2 Markdown Annotations:** Document each section with plain-language explanations of logic, formulas, variable choices, and statistical interpretations.

- **4.3 Visualization Formatting:** Finalize clear, presentation-grade plots (competition-tier box plots, distance-fare scatter plots, residual plots).
- **4.4 Run-Through Verification:** Run "Restart Kernel and Run All Cells" to ensure complete, reproducible execution without errors.

Phase 5: Presentation Slide Deck & Rehearsal (Deliverable 2)

- **5.1 Slide Content Development (8–10 Slides Max):**
    - *Slide 1:* Title, team members, project scope.
    - *Slide 2:* Problem context & the "Monopoly Tax" concept.
    - *Slide 3:* Pipeline architecture (Capture → Process → Store).
    - *Slide 4:* Key Finding 1 — Statistical markup across competition tiers (Box Plot).
    - *Slide 5:* Key Finding 2 — Short-haul vs. long-haul pricing dynamics (Scatter Plot).
    - *Slide 6:* Key Finding 3 — Fair-fare model & Top-10 gouged routes table.
    - *Slide 7:* Practical applications (Consumers, corporate travel managers, policy).
    - *Slide 8:* Limitations & future expansion.
- **5.2 Speaker Assignments:** Divide sections across all 5 team members so each person presents a clear 1.5- to 2-minute segment.
- **5.3 Timing & Tech Rehearsal:** Run timed rehearsals to stay under the 10-minute limit and prepare responses for the 2–3 minute Q&A. Verify display connectivity via HDMI/USB-C.